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Plasma processing apparatus

Abstract

A plasma processing apparatus comprises a processing chamber accommodating a substrate, and defining a processing space by an upper wall, a side wall, and a lower wall, a microwave generator configured to generate a microwave for generating plasma, a plurality of microwave radiators provided above the upper wall, and configured to radiate the microwave toward the processing chamber, a plurality of microwave transmission windows provided at positions corresponding to the plurality of microwave radiators in the upper wall, and formed of a dielectric, and a plurality of resonator array structures disposed on lower surfaces of the plurality of microwave transmission windows, respectively. The resonator array structures are formed by arranging a plurality of resonators that are capable of resonance with a magnetic field component of the microwave and are smaller in size than a wavelength of the microwave.

Inventors: **Kaneko; Kazushi (Yamanashi, JP), Kawakami; Satoru (Yamanashi, JP), Kamata; Eiki (Yamanashi, JP)**

Applicant: **TOKYO ELECTRON LIMITED** (Tokyo, JP)

Family ID: **91280237**

Assignee: **Tokyo Electron Limited (Tokyo, JP)**

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Primary Examiner: Crawford; Jason M

Attorney, Agent or Firm: Fenwick & West LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS
(1) This application claims priority to Japanese Patent Application No. 2022-192542, filed on Dec. 1, 2022, the entire contents of which are incorporated herein by reference.
TECHNICAL FIELD
(2) The present disclosure relates to a plasma processing apparatus.
BACKGROUND
(3) Japanese Laid-open Patent Publication No. 2021-031706 has described a plasma processing apparatus having a plurality of microwave radiation mechanisms on an upper wall of a processing chamber.
SUMMARY
(4) The present disclosure provides a plasma processing apparatus that can suppress interference between a plurality of microwave radiators.
(5) A plasma processing apparatus according to one aspect of the present disclosure, comprises: a processing chamber accommodating a substrate, and defining a processing space by an upper wall, a side wall, and a lower wall; a microwave generator configured to generate a microwave for generating plasma; a plurality of microwave radiators provided above the upper wall, and configured to radiate the microwave toward the processing chamber; a plurality of microwave transmission windows provided at positions corresponding to the plurality of microwave radiators in the upper wall, and formed of a dielectric; and a plurality of resonator array structures disposed on lower surfaces of the plurality of microwave transmission windows, respectively, and formed by arranging a plurality of resonators that are capable of resonance with a magnetic field component of the microwave and are smaller in size than a wavelength of the microwave.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a schematic cross-sectional view illustrating an example of the configuration of a plasma processing apparatus according to a first embodiment.
(2) FIG. **2** is a diagram illustrating an example of the configuration of a microwave introduction device according to the first embodiment.
(3) FIG. **3** is a diagram schematically illustrating an example of a microwave radiation mechanism according to the first embodiment.
(4) FIG. **4** is a diagram schematically illustrating an example of an upper wall of a processing chamber according to the first embodiment.
(5) FIG. **5** is a plan view illustrating an example of the configuration of a dielectric window and a resonator array structure according to the first embodiment when seen from below.
(6) FIG. **6** is a diagram illustrating an example of the configuration of a card-shaped resonator according to the first embodiment.
(7) FIG. **7** is a diagram illustrating an example of the configuration of the card-shaped resonator

according to the first embodiment.
(8) FIG. **8** is a diagram illustrating another example of the configuration of the card-shaped resonator according to the first embodiment.
(9) FIG. **9** is a diagram illustrating an example of a cross-section of the card-shaped resonator according to the first embodiment.
(10) FIG. **10** is a diagram illustrating an example of a relationship between the S21 value of the card-shaped resonator and the frequency of microwaves.
(11) FIG. **11** is a diagram illustrating an example of the arrangement of card-shaped resonators according to modification 1.
(12) FIG. **12** is a diagram illustrating an example of the arrangement of card-shaped resonators according to modification 2.
(13) FIG. **13** is a diagram illustrating an example of the arrangement of card-shaped resonators according to modification 3.
(14) FIG. **14** is a schematic cross-sectional view illustrating an example of the configuration of a plasma processing apparatus according to a second embodiment.
(15) FIG. **15** is a diagram schematically illustrating an example of an upper wall of a processing chamber according to the second embodiment.
(16) FIG. **16** is a schematic cross-sectional view illustrating an example of the configuration of a plasma processing apparatus according to a third embodiment.
(17) FIG. **17** is a diagram schematically illustrating an example of an upper wall of a processing chamber according to the third embodiment.
(18) FIG. **18** is a plan view illustrating an example of the configuration of a dielectric window and a resonator array structure according to the third embodiment when seen from below.
(19) FIG. **19** is a diagram illustrating an example of the arrangement of card-shaped resonators according to modification 4.
(20) FIG. **20** is a diagram illustrating an example of the arrangement of card-shaped resonators according to modification 5.
(21) FIG. **21** is a diagram illustrating an example of the arrangement of card-shaped resonators according to modification 6.
(22) FIG. **22** is a plan view illustrating an example of the configuration of a dielectric window and a resonator array structure according to a fourth embodiment when seen from below.
(23) FIG. **23** is a diagram illustrating an example of the configuration of a card-shaped resonator according to a fifth embodiment.
(24) FIG. **24** is a plan view illustrating an example of the configuration of a dielectric window and a resonator array structure according to the fifth embodiment when seen from below.
(25) FIG. **25** is a diagram illustrating an example of the arrangement of card-shaped resonators according to modification 7.
(26) FIG. **26** is a diagram illustrating an example of the arrangement of card-shaped resonators according to modification 8.

DETAILED DESCRIPTION

(27) Hereinafter, plasma processing apparatuses according to embodiments of the present disclosure will be described in detail with reference to the accompanying drawings. It should be noted that the present disclosure is not limited to the following embodiments.
(28) In a plasma processing apparatus using a microwave for exciting plasma, the power of the microwave supplied into a processing chamber may be increased to increase the electron density of the plasma. As the power of the microwave supplied into the processing chamber is increased, the electron density of the plasma may be increased.
(29) Here, it is known that the dielectric constant of space in the processing chamber becomes negative, when the electron density of the plasma reaches a specific upper limit by increasing the power of the microwave supplied into the processing chamber. This upper limit of the electron

density is appropriately called “cutoff density”. Further, a refractive index is known as an index indicating whether the microwave propagates through space. The refractive index N is expressed by the following Equation (1).

$N=\sqrt{\varepsilon}\sqrt{\mu}$ (1) where ε: dielectric constant, μ: permeability

(30) The permeability is generally positive. Thus, when the dielectric constant of the space in the processing chamber becomes negative, the refractive index of the space in the processing chamber becomes a pure imaginary number according to the above Equation (1). Thereby, the microwave is attenuated and may not propagate through the space in the processing chamber. As such, when the electron density of the plasma reaches the cutoff density, the power of the microwave is not sufficiently absorbed by the plasma because the microwave may not propagate through the space in the processing chamber. As a result, there is a problem in that increasing the density of plasma generated in the processing chamber over a wide range is inhibited.

(31) Further, in the plasma processing apparatus having a plurality of microwave radiators on an upper wall of the processing chamber, the plasma is generated directly below the upper wall. In such a plasma processing apparatus, when the high-frequency power of the microwave radiator is increased to generate high-density plasma, surface-wave plasma spreads to a neighboring microwave radiator, causing microwaves between the microwave radiators to interfere with each other. When the interference occurs, a plasma load fluctuates rapidly. Therefore, the plasma load may not be matched by a matcher, and a reflected wave is generated, making the plasma unstable. Further, when a reflected wave is generated, power consumed by the plasma decreases even if the power of a traveling wave is constant, so plasma processing speed decreases. Therefore, there is an expectation for a technology that can realize high density plasma over a wide range and suppress interference between the plurality of microwave radiators.

First Embodiment

(32) [Configuration of Plasma Processing Apparatus]

(33) FIG. **1** is a schematic cross-sectional view illustrating an example of the configuration of a plasma processing apparatus according to a first embodiment. The plasma processing apparatus **100** shown in FIG. **1** includes a processing chamber **101**, a mounting table **102**, a gas supply mechanism **103**, an exhaust device **104**, a microwave introduction device **105**, and a controller **106**. The processing chamber **101** accommodates a substrate W. The substrate W is mounted on the mounting table **102**. The gas supply mechanism **103** supplies gas into the processing chamber **101**. The exhaust device **104** exhausts the inside of the processing chamber **101**. The microwave introduction device **105** generates the microwave for generating plasma in the processing chamber **101**, and introduces the microwave into the processing chamber **101**. The controller **106** controls the operation of each component of the plasma processing apparatus **100**.

(34) The processing chamber **101** is formed of a metal material such as aluminum or its alloy, and defines a substantially cylindrical processing space S therein. The processing chamber **101** includes a plate-shaped upper wall **111** and lower wall **113**, and a side wall **112** connecting the upper and lower walls. The microwave introduction device **105** is provided on the processing chamber **101**, and functions as a plasma generating means that introduces an electromagnetic wave (microwave) into the processing chamber **101** to generate plasma. The microwave introduction device **105** will be described below in detail.

(35) The upper wall **111** includes a plurality of openings into which the microwave radiation mechanism, resonator array structure, and gas introduction portion of the microwave introduction device **105**, which will be described later, are inserted. The side wall **112** has a loading/unloading port **114** for loading and unloading the substrate W, which is a target to be processed, into and out from a transfer chamber (not shown) adjacent to the processing chamber **101**. The loading/unloading port **114** is opened and closed by a gate valve **115**. The lower wall **113** is provided with the exhaust device **104**. The exhaust device **104** is provided on an exhaust pipe **116**

connected to the lower wall **113**, and is provided with a vacuum pump and a pressure control valve. The inside of the processing chamber **101** is exhausted through the exhaust pipe **116** by the vacuum pump of the exhaust device **104**. The pressure in the processing chamber **101** is controlled by the pressure control valve.

(36) The mounting table **102** has a disk shape and is formed of ceramics such as AlN. The mounting table **102** is supported by a cylindrical support member **120** which is made of ceramics such as AlN and extends upward from the center of the lower portion of the processing chamber **101**. A guide ring **181** is provided on the outer periphery of the mounting table **102** to guide the substrate W. Further, a lifting pin (not shown) for raising and lowering the substrate W is provided in the mounting table **102** to move in and out of the upper surface of the mounting table **102**. In addition, a resistance heating type heater **182** is embedded in the mounting table **102**, and the heater **182** is supplied with power from a heater power supply **183** to heat the substrate W on the mounting table **102** via the mounting table. A thermocouple (not shown) is inserted into the mounting table **102**, and the heating temperature of the substrate W may be controlled to a predetermined temperature in the range of 300 to 1000° C., for example, on the basis of a signal from the thermocouple. Further, an electrode **184** of the same size as the substrate W is embedded above the heater **182** in the mounting table **102**, and a high-frequency bias power supply **122** is electrically connected to the electrode **184**. A high-frequency bias for attracting ions to the mounting table **102** from the high-frequency bias power supply **122** is applied to the electrode **184**. In addition, the high-frequency bias power supply **122** may not be provided depending on the plasma processing characteristics.

(37) The gas supply mechanism **103** serves to introduce a plasma generation gas and a raw gas for forming a target layer to be formed, such as a carbon layer, into the processing chamber **101**, and includes a plurality of gas introduction nozzles **123**. Each gas introduction nozzle **123** is inserted into the opening formed in the upper wall **111** of the processing chamber **101**. A gas supply pipe **191** is connected to each gas introduction nozzle **123**. This gas supply pipe **191** is branched into five branch pipes **191***a*, **191***b*, **191***c*, **191***d*, and **191***e*. An Ar gas source **192**, an O.sub.2 gas source **193**, a N.sub.2 gas source **194**, a H.sub.2 gas source **195**, and a C.sub.2H.sub.2 gas source **196** are connected to the branch pipes **191***a*, **191***b*, **191***c*, **191***d*, and **191***e*. The Ar gas source **192** supplies Ar gas as a rare gas that is a plasma generation gas. The O.sub.2 gas source **193** supplies O.sub.2 gas as an oxidizing gas that is a cleaning gas. The N.sub.2 gas source **194** supplies N.sub.2 gas used as a purge gas. The H.sub.2 gas source **195** supplies H.sub.2 gas as a reduction gas. The C.sub.2H.sub.2 gas source **196** supplies acetylene (C.sub.2H.sub.2) gas as a carbon containing gas that is a layer forming raw gas. Further, the C.sub.2H.sub.2 gas source **196** may supply another carbon containing gas such as ethylene (C.sub.2H.sub.4).

(38) Although not shown in the drawing, the branch pipes **191***a*, **191***b*, **191***c*, **191***d*, and **191***e* are provided with a mass flow controller for controlling a flow rate and valves provided before and after the mass flow controller. A shower plate may be further provided to supply C.sub.2H.sub.2 gas and H.sub.2 gas to a position closer to the substrate W, thus controlling the dissociation of gas. Further, the same effect can be obtained by extending the nozzles supplying these gases downward.

(39) As described above, the microwave introduction device **105** is provided above the processing chamber **101**, and functions as a plasma generating means that generates plasma by introducing the electromagnetic wave (microwave) into the processing chamber **101**.

(40) FIG. **2** is a diagram illustrating an example of the configuration of the microwave introduction device according to the first embodiment. As shown in FIGS. **1** and **2**, the microwave introduction device **105** includes the upper wall **111** of the processing chamber **101**, a microwave output portion **130**, and an antenna unit **140**. The upper wall **111** functions as an upper plate. The microwave output portion **130** generates the microwave, and distributes and outputs the microwave to a plurality of paths. The antenna unit **140** introduces the microwave output from the microwave output portion **130** into the processing chamber **101** through a resonator array structure **200** that

will be described later.
(41) The microwave output portion **130** includes a microwave power supply **131**, a microwave oscillator **132**, an amplifier **133**, and a distributor **134**. The microwave oscillator **132** is a solid state, and oscillates the microwave at, for example, 2.45 GHz (e.g., PLL oscillation). In addition, the frequency of the microwave is not limited to 2.45 GHz, and may be in the range of 700 MHz to 10 GHz, such as 860 MHz, 8.35 GHz, 5.8 GHz, or 1.98 GHz. The amplifier **133** amplifies the microwave oscillated by the microwave oscillator **132**. The distributor **134** distributes the microwave amplified by the amplifier **133** into a plurality of paths. The distributor **134** distributes the microwave while matching the impedance of an input side with the impedance of an output side.
(42) The microwave output portion **130** may adjust the frequency, power, and bandwidth of the microwave. The microwave output portion **130** may generate the microwave of a single frequency by setting the bandwidth of the microwave to approximately 0, for example. Further, the microwave output portion **130** may generate the microwave (hereinafter appropriately referred to as “wideband microwave”) containing a plurality of frequency components belonging to a predetermined frequency bandwidth. The plurality of frequency components may have the same power, and only a median frequency component within a band may have power greater than that of another frequency component. The microwave output portion **130** may adjust the power of the microwave within the range of 0 W to 5000 W, for example. The microwave output portion **130** may adjust the frequency of the microwave or the median frequency of the wideband microwave within the range of 2.3 GHz to 2.5 GHz, for example, and the bandwidth of the wideband microwave may be adjusted within the range of 0 MHz to 100 MHz, for example. In addition, the microwave output portion **130** may adjust the frequency pitch (carrier pitch) of the plurality of frequency components of the wideband microwave within the range of 0 to 25 kHz, for example. The microwave output portion **130** is an example of a microwave generator.
(43) The antenna unit **140** includes a plurality of antenna modules **141**. Each of the plurality of antenna modules **141** introduces the microwave distributed by the distributor **134** into the processing chamber **101**. The plurality of antenna modules **141** has the same configuration. Each antenna module **141** has an amplifier portion **142** that mainly amplifies and outputs the distributed microwave, and a microwave radiation mechanism **143** that radiates the microwave output from the amplifier portion **142** into the processing chamber **101**.
(44) The amplifier portion **142** includes a phase shifter **145**, a variable gain amplifier **146**, a main amplifier **147**, and an isolator **148**. The phase shifter **145** changes the phase of the microwave. The variable gain amplifier **146** adjusts the power level of the microwave input into the main amplifier **147**. The main amplifier **147** is formed as a solid state amplifier. The isolator **148** separates reflected microwaves that are reflected by an antenna portion of the microwave radiation mechanism **143**, which will be described later, and are directed toward the main amplifier **147**.
(45) Here, the microwave radiation mechanism **143** will be described with reference to FIG. **3**. FIG. **3** is a diagram schematically illustrating an example of the microwave radiation mechanism according to the first embodiment. As shown in FIG. **1**, the plurality of microwave radiation mechanisms **143** are provided on the upper wall **111**. Further, as shown in FIG. **3**, the microwave radiation mechanism **143** includes a cylindrical outer conductor **152** and an inner conductor **153** that is provided inside the outer conductor **152** to be coaxial with the outer conductor **152**. The microwave radiation mechanism **143** includes a coaxial tube **151** having a microwave transmission line between the outer conductor **152** and the inner conductor **153**, a tuner **154**, a power feeder **155**, and an antenna portion **156**. The tuner **154** matches the impedance of the load with the characteristic impedance of the microwave power supply **131**. The power feeder **155** feeds the microwave amplified from the amplifier portion **142** to the microwave transmission line. The antenna portion **156** radiates the microwave from the coaxial tube **151** into the processing chamber **101**. Further, the microwave radiation mechanism **143** is an example of the microwave radiator.

(46) The power feeder **155** introduces the microwave amplified in the amplifier portion **142** by a coaxial cable from the side of the upper end of the outer conductor **152**, and radiates the microwave using a feeding antenna, for example. By radiating the microwave, the microwave power is fed to the microwave transmission line between the outer conductor **152** and the inner conductor **153**, and the microwave power propagates toward the antenna portion **156**.
(47) The antenna portion **156** is provided on the lower end of the coaxial tube **151**. The antenna portion **156** includes a disk-shaped planar antenna **161** connected to the lower end of the inner conductor **153**, a slow wave material **162** disposed on the upper surface of the planar antenna **161**, and a microwave transmitting plate **163** disposed on the lower surface of the planar antenna **161**. The microwave transmitting plate **163** is an example of a microwave transmission window. The microwave transmitting plate **163** is fitted into the upper wall **111**, and a lower surface thereof faces the resonator array structure **200**, which will be described later, and is exposed to the internal space of the processing chamber **101**. The planar antenna **161** has a slot **161***a* formed to penetrate therethrough. The shape of the slot **161***a* is appropriately set so that the microwave is efficiently radiated. A dielectric may be inserted into the slot **161***a*.
(48) The slow wave material **162** is formed of a material having a dielectric constant larger than that of vacuum, and allows the phase of the microwave to be adjusted using a thickness thereof, thus maximizing the radiation energy of the microwave. The microwave transmitting plate **163** is also made of a dielectric material, and has a shape that allows the microwave to be efficiently radiated in a TE mode. The microwave transmitted through the microwave transmitting plate **163** generates plasma in space defined in the processing chamber **101** via the resonator array structure **200**, which will be described later. As the material forming the slow wave material **162** and the microwave transmitting plate **163**, for example, quartz, ceramics, fluorine-based resin such as polytetrafluoroethylene resin, polyimide resin, etc. may be used.
(49) The tuner **154** constitutes a slug tuner. As shown in FIG. **3**, the tuner **154** includes slugs **171***a* and **171***b*, an actuator **172**, and a tuner controller **173**. The slugs **171***a* and **171***b* are two slugs disposed on a base end side (upper end side) rather than the antenna portion **156** of the coaxial tube **151**. The actuator **172** independently drives the two slugs. A tuner controller **173** controls the actuator **172**.
(50) The slugs **171***a* and **171***b* are plate-shaped and ring-shaped, are made of a dielectric material such as ceramics, and are arranged between the outer conductor **152** and the inner conductor **153** of the coaxial tube **151**. For example, the actuator **172** individually drives the slugs **171***a* and **171***b* by rotating two screws provided inside the inner conductor **153** and fastened to the slugs **171***a* and **171***b*. On the basis of a command from the tuner controller **173**, the slugs **171***a* and **171***b* are moved vertically by the actuator **172**. The tuner controller **173** adjusts the positions of the slugs **171***a* and **171***b* so that the impedance at a terminal end becomes 50Ω.
(51) The main amplifier **147**, the tuner **154**, and the planar antenna **161** are arranged to be close to each other. The tuner **154** and the planar antenna **161** constitute a lumped constant circuit, and also functions as a resonator. Impedance mismatch exists in the attachment part of the planar antenna **161**. However, since the tuner **154** directly tunes the plasma load, tuning including plasma may be performed with high precision, and the influence of reflection on the planar antenna **161** may be eliminated.
(52) FIG. **4** is a diagram schematically illustrating an example of the upper wall of a processing chamber according to the first embodiment. FIG. **4** corresponds to a portion of the processing space S inside the side wall **112** in the cross-section taken along line IV-IV of FIG. **1**. As shown in FIG. **4**, according to the first embodiment, seven microwave radiation mechanisms **143** are provided, and the corresponding microwave transmitting plates **163** are evenly arranged in a hexagonal close-packed structure. In other words, one of the seven microwave transmitting plates **163** is arranged at the center of the upper wall **111**, and six microwave transmitting plates **163** are arranged around the central plate. The seven microwave transmitting plates **163** are arranged so that adjacent

microwave transmitting plates **163** are spaced apart from each other at regular intervals. The plurality of gas introduction nozzles **123** of the gas supply mechanism **103** are arranged to surround the central microwave transmitting plate **163**. Further, the number of the microwave radiation mechanisms **143** is not limited to seven.

(53) The resonator array structure **200** is provided on the lower surface of the microwave transmitting plate **163**, in other words, at a position corresponding to each of the plurality of microwave radiation mechanisms **143** on the upper wall **111**. The resonator array structure **200** is formed by arranging a plurality of resonators that are capable of resonance with the magnetic field component of the microwave and are smaller in size than the wavelength of the microwave, and is located in the processing chamber **101**. The lower surface of the microwave transmitting plate **163** may be in contact with or spaced apart from the resonator array structure **200**.

(54) By locating the resonator array structure **200** in the processing chamber **101**, it is possible to cause the microwave supplied to the processing space S by the microwave radiation mechanism **143** to resonate with the plurality of resonators of the resonator array structure **200**. By the resonance of the microwave and the plurality of resonators, the microwave may be efficiently supplied to the processing space S of the processing chamber **101**, and the permeability of the processing space S may be made negative. When the permeability of the processing space S is negative, the electron density of the plasma generated in the processing space S may reach the cutoff density, and the dielectric constant of the processing space S may be negative. Even in this case, since the refractive index becomes a real number according to the above Equation (1), the microwave may propagate in the processing space S. Thus, even if the electron density of the plasma generated in the processing space S reaches the cutoff density, the microwave may propagate beyond the skin depth of the plasma, and the microwave power is efficiently absorbed by the plasma. As a result, the high-density plasma may be generated over a wide range beyond the skin depth of the plasma. That is, the plasma processing apparatus **100** according to the present embodiment can increase the density of the plasma over a wide range by locating the resonator array structure **200** in the processing chamber **101**.

(55) Here, the detailed configuration of the resonator array structure **200** will be described with reference to FIGS. **1** and **5**. FIG. **5** is a plan view illustrating an example of the configuration of the dielectric window and the resonator array structure according to the first embodiment when seen from below. In FIG. **5**, one of the plurality of microwave transmitting plates **163** that are dielectric windows has a disk-shaped lower surface.

(56) The resonator array structure **200** is formed by arranging in a grid shape a plurality of resonators **201** which are capable of resonance with the magnetic field component of the microwave and are smaller in size than the wavelength of the microwave. To be more specific, the plurality of resonators **201** includes at least one resonator among resonators **201**A and **201**B shown in FIGS. **6** and **7**. Each of the plurality of resonators **201** forms a series resonant circuit composed of a capacitor equivalent element and a coil equivalent element. The series resonant circuit is realized by patterning a conductor on a plane. Further, each of the plurality of resonators **201** has a size less than 1/10 of the wavelength of the microwave.

(57) FIG. **6** is a diagram illustrating an example of the configuration of a card-shaped resonator according to the first embodiment. The card-shaped resonator **201**A shown in FIG. **6** has a structure in which two C-shaped ring members **211**A are stacked on one surface of a dielectric plate **212**A. The two C-shaped ring members **211**A are made of conductors and concentric circles arranged in opposite directions. The capacitor equivalent elements are formed on the opposing surfaces of the inner ring member **211**A and the outer ring member **211**A or both ends of each ring member **211**A, and the coil equivalent element is formed along each ring member **211**A. Thereby, the resonator **201**A may form a series resonant circuit.

(58) FIG. **7** is a diagram illustrating an example of the configuration of the card-shaped resonator according to the first embodiment. The card-shaped resonator **201**B shown in FIG. **7** has a structure

in which two C-shaped ring members **211**B made of conductors are arranged, and a dielectric plate **212**B is interposed between the ring members **211**B arranged adjacent to each other in opposite directions. That is, in the resonator **201**B, the dielectric plate **212**B is inserted between two C-shaped ring members **211**B facing in opposite directions. The capacitor equivalent elements are formed on the opposing surfaces of the two C-shaped ring members **211**B or both ends of each ring member **211**B, and the coil equivalent element is formed along each ring member **211**B. Thereby, the resonator **201**B may form a series resonant circuit. The card-shaped resonator **201**B may be expressed as being formed into a card shape for each set of two C-shaped ring members **211**B.
(59) In the resonator **201**B shown in FIG. **7**, the arranged number of ring members **211**B (hereinafter referred to as a “stacked number”) is two, but the stacked number of the ring members **211**B may be greater than two. FIG. **8** is a diagram illustrating another example of the configuration of the card-shaped resonator according to the first embodiment. The resonator **201**B shown in FIG. **8** has a structure in which N (N≥2) C-shaped ring members **211**B made of conductors are arranged and a dielectric plate **212**B is interposed between the ring members **211**B arranged adjacent to each other in opposite directions. Even with such a structure, the resonator **201**B may form a series resonant circuit.
(60) Further, an insulating film may be formed on each of the plurality of resonators **201**. FIG. **9** is a diagram illustrating an example of a cross-section of the card-shaped resonator according to the first embodiment. FIG. **9** shows a side cross-section of the resonator **201**B shown in FIG. **7**. An insulating film **213** (an example of the dielectric layer) is formed on the surface of the resonator **201**B. The material of the film **213** is, for example, ceramic. The thickness of the film **213** is in the range of 0.001 mm to 2 mm, for example. The insulating film **213** may be formed on each of the plurality of resonators **201**, thus suppressing abnormal discharge in each of the plurality of resonators **201**. Further, the insulating film **213** may be formed on each of the plurality of resonators **201**, thus suppressing the exposure of the ring member **211**B to plasma.
(61) The controller **106** includes a processor, a memory, and an input/output interface. The memory stores a program, a process recipe, etc. The processor reads out the program from the memory and executes the program to centrally control each component of the plasma processing apparatus **100** though the input/output interface on the basis of the process recipe stored in the memory.
(62) For example, when plasma is generated in the processing space S, the controller **106** controls such that the microwave supplied to the processing space S by the microwave radiation mechanism **143** resonates with the plurality of resonators **201** in a target frequency band higher than the resonant frequency of the plurality of resonators **201**. For example, the resonant frequency is a frequency at which the transmission characteristic value (e.g., S21 value) of the plurality of resonators **201** becomes a minimum value.
(63) FIG. **10** is a diagram illustrating an example of a relationship between the S21 value of the card-shaped resonator and the frequency of the microwave. When the frequency of the microwave supplied to the processing space S by the microwave radiation mechanism **143** matches the resonant frequency fr (=about 2.35 GHz) of the plurality of resonators **201**, the S21 value of the plurality of resonators **201** becomes the minimum value, and resonance between the microwave and the plurality of resonators **201** occurs. The resonance between the microwave and the plurality of resonators **201** is maintained even in a predetermined frequency band (e.g., about 0.1 GHz) higher than the resonant frequency fr of the plurality of resonators **201**. In a predetermined frequency band higher than the resonant frequency fr of the plurality of resonators **201**, both the dielectric constant and the permeability of the processing space S may be made negative due to the resonance between the microwave and the plurality of resonators **201**. As can be seen from the above Equation (1), the microwave may propagate in the processing space S. The target frequency band in this embodiment is set to a predetermined frequency band (e.g., about 0.1 GHz) higher than the resonant frequency fr of the plurality of resonators **201**. For example, the target frequency band is preferably within 0.05 times the resonant frequency fr of the plurality of resonators **201**. It

should be noted that the resonant frequency fr of the card-shaped resonator **201** can be measured using, for example, a vector network analyzer in which a transmission and reception antenna is arranged in a direction parallel to the ring member **211**B of the card-shaped resonator **201**B.

(64) Regarding the propagation of the electromagnetic wave to the plurality of resonators, the relationship between the resonant frequency, the refractive index, the dielectric constant, and the permeability has been reported by, for example, “Electromagnetic parameter retrieval from inhomogeneous metamaterials” in “PHYSICAL REVIEW E 71, 036617 (2005)” by D. R. Smith, D. C. Vier, Th. Koschny and C. M. Soukoulis.

(65) In this way, even if the electron density of the plasma reaches the cutoff density by causing the microwave and the plurality of resonators **201** to resonate in the target frequency band higher than the resonant frequency fr of the plurality of resonators **201**, the microwave may propagate beyond the skin depth of the plasma. Thus, the plasma may efficiently absorb the power of the microwave. As a result, the high-density plasma may be generated over a wide range beyond the skin depth of the plasma. That is, in the plasma processing apparatus **100** according to the present embodiment, the high-density plasma can be realized over a wide range by causing the microwave to resonate with the plurality of resonators **201** in a target frequency band higher than the resonant frequency fr of the plurality of resonators **201**.

(66) [Arrangement of Multiple Resonators]

(67) Next, the arrangement of the plurality of resonators **201** in the resonator array structure **200** will be described. As shown in FIG. **5**, in the resonator array structure **200**, the card-shaped resonators **201** are arranged in a grid shape. In the following description, the resonator array structure **200** may be referred to as a metamaterial **200**, and the card-shaped resonator **201** may be referred to as a meta-atom **201**. In the example of FIG. **5**, meta-atoms **201** are arranged so that cells **220** surrounded by the meta-atoms **201** are formed in five columns in an X-axis direction and in five rows in a Y-axis direction. In other words, the cells **220** are arranged in a 5×5 square array **230**. In the array **230**, the diameter of the microwave transmitting plate **163** and the length of one side of the metamaterial **200** are substantially equal to each other. Therefore, except for some cell **220** (1st row and 3rd column, 3rd row and 1st column, 3rd row and 5th column, 5th row and 3rd column), the cells **220** in a peripheral portion are arranged across the microwave transmitting plate **163** and the upper wall **111**. As such, when some cells **220** of the resonator array structure **200** are arranged across the microwave transmitting plate **163** and the upper wall **111**, the upper wall **111** is preferably a dielectric so as to propagate the microwave.

(68) According to this embodiment, it is possible to prevent the plasma from spreading in the horizontal direction from the metamaterial **200** by disposing the metamaterial **200** on the lower surface of the microwave transmitting plate **163**. In other words, since the plasma is confined in the cell **220** of the metamaterial **200**, interference between neighboring microwave radiation mechanisms **143** may be suppressed. Further, it is possible to suppress the plasma from spreading to the side wall **112** or the gas introduction nozzle **123**. Therefore, contamination of aluminum, yttrium, etc. due to damage to the side wall **112** or abnormal discharge near the gas introduction nozzle **123** may be suppressed. Further, a process processing speed can be stabilized by stabilizing plasma discharge. Although not shown in the drawing, the plurality of meta-atoms **201** may be arranged on a base plate formed of a dielectric material, and may be the metamaterial **200** including the base plate. In this case, the metamaterial **200** may be easily attached to the upper wall **111**.

(69) [Modifications 1 to 3 of Resonator Array Structure]

(70) Here, modifications in the arrangement of the meta-atom **201** will be described as modifications 1 to 3 in the arrangement of the resonator array structure (metamaterial) **200**. FIG. **11** is a diagram illustrating an example of the arrangement of card-shaped resonators according to modification 1. The metamaterial **200**A shown in FIG. **11** has an array **231** obtained by removing four corner cells **220** from the array **230** shown in FIG. **5**. In the array **231**, a portion of the microwave transmitting plate **163** is located outside the metamaterial **200**A. Further, in the array

231, some of the cells **220** of the peripheral portion (1st row and 2nd column, 1st row and 4th column, 2nd row and 1st column, 2nd row and 5th column, 4th row and 1st column, 4th row and 5th column, 5th row and 2nd column, and 5th row and 4th column) are arranged across the microwave transmitting plate **163** and the upper wall **111**.

(71) FIG. **12** is a diagram illustrating an example of the arrangement of card-shaped resonators according to modification 2. The metamaterial **200**B shown in FIG. **12** has an array **232** obtained by further removing some of the cells **220** of the peripheral portion (1st row and 2nd column, 1st row and 4th column, 2nd row and 1st column, 2nd row and 5th column, 4th row and 1st column, 4th row and 5th column, 5th row and 2nd column, and 5th row and 4th column) from the array **231** shown in FIG. **11**. In the array **231**, a portion of the microwave transmitting plate **163** is located outside the metamaterial **200**B. In the array **232**, there is no cell **220** arranged across the microwave transmitting plate **163** and the upper wall **111**.

(72) FIG. **13** is a diagram illustrating an example of the arrangement of card-shaped resonators according to modification 3. The metamaterial **200**C shown in FIG. **13** has an array **233** in which the outermost meta-atom **201** of the array **230** shown in FIG. **5** is replaced with a (non-resonant) meta-atom **201**C having a resonant frequency fr different from that of the meta-atom **201**. In the array **233**, for example, the meta-atom **201** has a resonant frequency fr (=about 2.35 GHz). Further, for example, the meta-atom **201**C has a resonant frequency fr (=about 2.15 GHz or about 2.55 GHz) that is a predetermined frequency (e.g., ±0.2 GHz) away from the resonant frequency fr of the meta-atom **201**. In the array **233**, there are a cell **221** surrounded on all sides by the meta-atoms **201** and a cell **222** surrounded on one or two sides by the meta-atoms **201**C. Since the number of meta-atoms **201** that resonate with the frequency output from the microwave output portion **130** in the cell **222** is smaller than that in the cell **221**, the plasma density in the cell **222** is lower than that in the cell **221**, and it is possible to suppress plasma from spreading to the outside of the metamaterial **200**C.

Second Embodiment

(73) In the above-described first embodiment, the individual metamaterial **200** is placed for each of the plurality of microwave transmitting plates **163**. However, the plurality of metamaterials **200** may be integrated as one upper plate. This will be described as a second embodiment. Since the plasma processing apparatus of the second embodiment is the same as that of the above-described first embodiment except for the configuration of the upper plate in which the metamaterial **200** is integrated, a duplicated description of the configuration and operation will be omitted.

(74) FIG. **14** is a schematic cross-sectional view illustrating an example of the configuration of a plasma processing apparatus according to a second embodiment. FIG. **15** is a diagram schematically illustrating an example of an upper wall of a processing chamber according to the second embodiment. FIG. **15** corresponds to a portion of the processing space S inside the side wall **112** in the cross-section taken along line XV-XV of FIG. **14**. As shown in FIGS. **14** and **15**, the resonator array structure **200** is provided on the lower surface of the microwave transmitting plate **163**, that is, at a position corresponding to each of the plurality of microwave radiation mechanisms **143** on the upper wall **111**. Each resonator array structure **200** is embedded and integrated into the upper plate **240**. The upper plate **240** is formed of a dielectric (insulator) such as quartz or alumina. Further, a hole is provided in the upper plate **240** at a position corresponding to the gas introduction nozzle **123** so that gas may be introduced into the processing space S. In this way, each resonator array structure **200** may be easily installed on the upper wall **111** by forming each resonator array structure **200** integrally with the upper plate **240**.

Third Embodiment

(75) In the above-described first embodiment, the cells **220** of the peripheral portion are arranged across the microwave transmitting plate **163** and the upper wall **111**. However, all cells may be located in a range corresponding to the microwave transmitting plate **163**. This will be described as a third embodiment. Since the plasma processing apparatus of the third embodiment is the same as

that of the above-described first embodiment except for the size and arrangement of the metamaterial, a duplicated description of the configuration and operation will be omitted.
(76) FIG. **16** is a schematic cross-sectional view illustrating an example of the configuration of a plasma processing apparatus according to a third embodiment. FIG. **17** is a diagram schematically illustrating an example of an upper wall of a processing chamber according to the third embodiment. FIG. **17** corresponds to a portion of the processing space S inside the side wall **112** in the cross-section taken along line XVII-XVII of FIG. **16**. As shown in FIGS. **16** and **17**, the resonator array structure (metamaterial) **200**D is provided on the lower surface of the microwave transmitting plate **163**, that is, at a position corresponding to each of the plurality of microwave radiation mechanisms **143** on the upper wall **111**.
(77) FIG. **18** is a plan view illustrating an example of the configuration of a dielectric window and a resonator array structure according to the third embodiment when seen from below. In FIG. **18**, the lower surface of one microwave transmitting plate **163** among the plurality of microwave transmitting plates **163**, which are dielectric windows, is shown in a disk shape. As shown in FIG. **18**, the meta-atoms **201**D are arranged in the resonator array structure **200**D such that the cells **250** surrounded by the meta-atoms **201**D are formed in five columns in the X-axis direction and five rows in the Y-axis direction. In other words, the cells **250** are arranged in a 5×5 square array **260**. In the array **260**, the length of the diagonal line of the rectangle of the metamaterial **200**D is equal to or less than the diameter of the microwave transmitting plate **163**. Therefore, all the cells **250** are within the range of the microwave transmitting plate **163**. In this way, when all the cells **250** of the resonator array structure **200**D are within the range of the microwave transmitting plate **163**, the upper wall **111** may be a conductor of aluminum or the like. Furthermore, the meta-atom **201**D is smaller in size than the meta-atom **201** of the first embodiment.
(78) [Modifications 4 to 6 of Resonator Array Structure]
(79) Here, modifications in the arrangement of the meta-atom **201**D will be described as modifications 4 to 6 in the arrangement of the resonator array structure (metamaterial) **200**. FIG. **19** is a diagram illustrating an example of the arrangement of card-shaped resonators according to modification 4. The metamaterial **200**E shown in FIG. **19** has an array **261** obtained by removing four corner cells **250** from the array **260** shown in FIG. **18**. Since there are no cells **250** at four corners that are closet to the outer periphery of the microwave transmitting plate **163** in the array **261**, the plasma density between the cells **250** may be made uniform.
(80) FIG. **20** is a diagram illustrating an example of the arrangement of card-shaped resonators according to modification 5. The metamaterial **200**F shown in FIG. **20** has an array **262** obtained by further removing some of the cells **250** of the peripheral portion (1st row and 2nd column, 1st row and 4th column, 2nd row and 1st column, 2nd row and 5th column, 4th row and 1st column, 4th row and 5th column, 5th row and 2nd column, and 5th row and 4th column) from the array **261** shown in FIG. **19**. Since the array **262** has no cells **250** closer to the outer periphery of the microwave transmitting plate **163** than in the array **261**, the plasma density between the cells **250** may be made more uniform than in the array **261**.
(81) FIG. **21** is a diagram illustrating an example of the arrangement of card-shaped resonators according to modification 6. The metamaterial **200**E shown in FIG. **21** has an array **263** in which the outermost meta-atom **201**D of the array **260** shown in FIG. **18** is replaced with a meta-atom **201**E (non-resonant) having a resonant frequency fr different from that of the meta-atom **201**D. In the array **263**, for example, the meta-atom **201**D has a resonant frequency fr (=about 2.35 GHz). Further, for example, the meta-atom **201**E has a resonant frequency fr (=about 2.15 GHz or about 2.55 GHz) that is a predetermined frequency (e.g., ±0.2 GHz) away from the resonant frequency fr of the meta-atom **201**D. In the array **263**, there are a cell **251** surrounded on all sides by the meta-atoms **201**D and a cell **252** surrounded on one or two sides by the meta-atoms **201**E. Since the number of meta-atoms **201**D in the cell **252** that resonate with the frequency output from the microwave output portion **130** is smaller than that in the cell **251**, the plasma density in the cell **252**

is lower than that in the cell **251**.

Fourth Embodiment

(82) In the above-described modifications 3 and 6, the outermost meta-atoms of the metamaterials **200**C and **200**E are replaced with meta-atoms **201**C and **201**E that are non-resonant with respect to the frequency output from the microwave output portion **130**. In the plasma processing apparatus where the gas introduction nozzle is located in the center cell of the metamaterial, the center cell may be formed of non-resonant meta-atoms, thus preventing the plasma from spreading to the gas introduction nozzle. This will be described as a fourth embodiment. Since the plasma processing apparatus of the fourth embodiment is the same as that of the above-described first embodiment except that there is only one microwave radiation mechanism **143** and the gas introduction nozzle is provided in the center of the upper wall **111**, a duplicated description of the configuration and operation will be omitted.

(83) FIG. **22** is a plan view illustrating an example of the configuration of a dielectric window and a resonator array structure according to a fourth embodiment when seen from below. In FIG. **22**, the lower surface of the microwave transmitting plate **163**A, which is the dielectric window and occupies most of the upper wall **111**A, has a disk shape. The gas introduction nozzle **123**A is located at the center of the microwave transmitting plate **163**A. As shown in FIG. **22**, in the resonator array structure (metamaterial) **200**F, cells **253** surrounded by meta-atoms **201**F are formed in five columns in the X-axis direction and in five rows in the Y-axis direction. Further, in the metamaterial **200**F, the gas introduction nozzle **123**A is disposed, and the cell **254** surrounded by a non-resonant meta-atom **201**G is formed at the center. In other words, the cells **253** and **254** are arranged in a 5×5 square array **265**. In the array **265**, the length of the diagonal of the rectangle of the metamaterial **200**F is equal to or less than the diameter of the microwave transmitting plate **163**A. Therefore, all the cells **253** and **254** are within the range of the microwave transmitting plate **163**A.

(84) According to the fourth embodiment, in each cell **253**, the meta-atom **201**F resonates with the frequency output from the microwave output portion **130** to generate plasma. However, since the cell **254** is surrounded by meta-atoms **201**G that do not resonate with the frequency output from the microwave output portion **130**, plasma is not generated. Further, since the plasma generated in each cell **253** is confined in each cell **220**, it is possible to suppress the plasma from spreading to the inside of the cell **254** and the outside of the metamaterial **200**F in a horizontal direction. That is, in the fourth embodiment, plasma generation may be suppressed near the gas introduction nozzle **123**A and near the side wall. Although it is described in the fourth embodiment that the plasma processing apparatus has one microwave radiation mechanism **143**, the present disclosure may be applied to a plasma processing apparatus which has a plurality of microwave radiation mechanisms **143** and in which a gas introduction nozzle **123**A is disposed at the center of each microwave transmitting plate **163**.

Fifth Embodiment

(85) In the above-described first embodiment, the cells **220** are divided by the meta-atom **201**. However, the meta-atom may be provided with a through hole to ensure the continuity of plasma between neighboring cells. This will be described as the fifth embodiment. Since the plasma processing apparatus of the fifth embodiment remains the same as that of the above-described first embodiment except for the through hole of the metamaterial, a duplicated description of the configuration and operation will be omitted.

(86) FIG. **23** is a diagram illustrating an example of the configuration of a card-shaped resonator according to the fifth embodiment. The card-shaped resonator (meta-atom) **201**H shown in FIG. **23** has a structure in which two C-shaped ring members **211**H made of conductors are arranged, and a dielectric plate **212**H is interposed between the ring members **211**H arranged adjacent to each other in opposite directions. That is, in the meta-atom **201**H, the dielectric plate **212**H is inserted between two C-shaped ring members **211**H facing in opposite directions. The capacitor equivalent elements

are formed on the opposing surfaces of the two C-shaped ring members **211**H or both ends of each ring member **211**H, and the coil equivalent element is formed along each ring member **211**H. Thereby, the meta-atom **201**H may form a series resonant circuit. Further, the meta-atom **201**H has on the center a through hole **214**. By providing the through hole **214**, gas or plasma may flow in and out of the cell formed of the meta-atom **201**H between adjacent cells. The card-shaped resonator **201**H may be expressed as being formed into a card shape for each set of two C-shaped ring members **211**H.

(87) FIG. **24** is a plan view illustrating an example of the configuration of a dielectric window and a resonator array structure according to the fifth embodiment when seen from below. In FIG. **24**, the lower surface of one of a plurality of microwave transmitting plates **163**, which are dielectric windows, has a disk shape. As shown in FIG. **24**, a resonator array structure (metamaterial) **200**G has on an outer periphery thereof the meta-atom **201** that is the same as the first embodiment, with a meta-atom **201**H arranged therein. In the metamaterial **200**G, cells **223** surrounded by the meta-atoms **201** and **201**H are formed in five columns in the X-axis direction and in five rows in the Y-axis direction. In other words, the cells **223** are arranged in a 5×5 square array **270**. In the array **270**, the diameter of the microwave transmitting plate **163** is substantially equal to the length of one side of the metamaterial **200**G. Therefore, the cells **223** of the peripheral portion are arranged across the microwave transmitting plate **163** and the upper wall **111** except for some cells **223** (1st row and 3rd column, 3rd row and 1st column, 3rd row and 5th column, 5th row and 3rd column).

(88) In the first embodiment, when the meta-atom **201** resonates with the frequency output from the microwave output portion **130** and plasma ignites, the shape of a magnetic field entering each cell **220** is changed depending on the temporal change of the microwave or the timing of a pulse, so a cell **220** that does not ignite may occur. Once the state is stabilized after the plasma ignites, it is intended to be maintained as it is. Thus, cells that are not ignited (discharged) may remain in an unignited state. The unignited cells may degrade the uniformity of plasma processing.

(89) On the other hand, in the array **270** of the fifth embodiment, the plasma spreads from the cell **223** where the plasma is ignited through the through hole **214** to an adjacent cell **223**, so that the plasma is easily ignited even in the unignited cell **223**. In other words, in the array **270**, the unignited cell **223** is ignited in a plasma-assisted manner (chain reaction) using the plasma of the ignited cell **223**. Since meta-atoms **201** having no through hole **214** are arranged on the outermost periphery of the metamaterial **200**G, it is possible to suppress the plasma from spreading horizontally from the metamaterial **200**G. Further, in the fifth embodiment, it is possible to improve the uniformity of plasma of each cell **223**. In other words, according to the fifth embodiment, the high-density plasma can be generated and in-plane uniformity with respect to the substrate W can be improved. In addition, by combining the array of the meta-atoms **201** and **201**H, the plasma density may be controlled to be changed between central and peripheral portions.

(90) [Modifications 7 and 8 of Resonator Array Structure]

(91) Here, as modifications 7 and 8 of the array of the resonator array structure (metamaterial) **200**G, variations in the array of the meta-atoms **201** and **201**H will be described. FIG. **25** is a diagram illustrating an example of the arrangement of card-shaped resonators according to modification 7. The metamaterial **200**H shown in FIG. **25** has an array **271** obtained by removing four corner cells **223** from the array **270** shown in FIG. **24**. In the array **271**, the new outermost metal-atom **201**H in some of the cells **223** of the peripheral portion (1st row and 2nd column, 1st row and 4th column, 2nd row and 1st column, 2nd row and 5th column, 4th row and 1st column, 4th row and 5th column, 5th row and 2nd column, and 5th row and 4th column) is changed to the meta-atom **201** having no through hole **214**. Further, in the array **271**, a portion of the microwave transmitting plate **163** is located outside the metamaterial **200**H. Further, in the array **271**, some of the cells **223** of the peripheral portion (1st row and 2nd column, 1st row and 4th column, 2nd row and 1st column, 2nd row and 5th column, 4th row and 1st column, 4th row and 5th column, 5th row and 2nd column, and 5th row and 4th column) are arranged across the microwave transmitting

plate **163** and the upper wall **111**.
(92) FIG. **26** is a diagram illustrating an example of the arrangement of card-shaped resonators according to modification 8. The metamaterial **200**I shown in FIG. **26** has an array **272** obtained by further removing some of the cells **223** of the peripheral portion (1st row and 2nd column, 1st row and 4th column, 2nd row and 1st column, 2nd row and 5th column, 4th row and 1st column, 4th row and 5th column, 5th row and 2nd column, and 5th row and 4th column) from the array **271** shown in FIG. **25**. In the array **272**, the new outermost metal-atom **201**H in the cells **223** of the peripheral portion (1st row and 3rd column, 2nd row and 2nd column, 2nd row and 4th column, 3rd row and 1st column, 3rd row and 5th column, 4th row and 2nd column, 4th row and 4th column, and 5th row and 3rd column) is changed to the meta-atom **201** having no through hole **214**. In the array **272**, a portion of the microwave transmitting plate **163** is located outside the metamaterial **200**I. In the array **272**, there is no cell **223** arranged across the microwave transmitting plate **163** and the upper wall **111**. Thus, the arrangement of the meta-atoms **201** and **201**H may be changed depending on, for example, the high-frequency power for generating plasma.
(93) (Other Modifications)
(94) Further, in each embodiment, the metamaterials **200**, **200**A to **200**I are formed by arranging in a grid shape the plurality of meta-atoms **201**, **201**C to **201**H that are capable of resonance with the magnetic field component of the microwave and are smaller in size than the wavelength of the microwave. This case has been described as an example. The arrangement of the plurality of meta-atoms **201**, **201**C to **201**H is not limited thereto, and may be any arrangement such as a triangular or hexagonal arrangement. Further, for example, the plurality of meta-atoms **201**, **201**C to **201**H may be arranged at predetermined intervals along one direction. Further, the meta-atoms may be horizontally long rectangular meta-atoms in which the plurality of meta-atoms **201**, **201**C to **201**H are connected horizontally. For example, by combining the rectangular meta-atoms vertically and horizontally, a grid-like cell may be formed.
(95) As described above, according to each embodiment, the plasma processing apparatus **100** includes the processing chamber **101**, the microwave generator (microwave output portion **130**), the plurality of microwave radiators (microwave radiation mechanism **143**), the plurality of microwave transmission windows (microwave transmitting plate **163**), and the plurality of resonator array structures (metamaterial) **200**. The processing chamber **101** accommodates the substrate W, and defines the processing space S by the upper wall **111**, the side wall **112**, and the lower wall **113**. The microwave generator is configured to generate the microwave for generating the plasma. The microwave radiator is provided above the upper wall **111**, and is configured to radiate the microwave toward the processing chamber **101**. The microwave transmission window is provided at a position corresponding to each of the plurality of microwave radiators on the upper wall **111**, and is the microwave transmission window formed of the dielectric. The resonator array structure **200** is disposed on the lower surface of each of the plurality of microwave transmission windows, and is formed by arranging the plurality of resonators (meta-atoms **201**) that are capable of resonance with the magnetic field component of the microwave and are smaller in size than the wavelength of the microwave. As a result, interference between the plurality of microwave radiators can be suppressed. Further, it is possible to suppress damage to the gas introduction nozzle **123** or the side wall **112** due to the plasma.
(96) Further, according to the first, third to fifth embodiments, the plurality of resonator array structures **200**, **200**A to **200**I are individually formed. As a result, each of the resonator array structures **200**, **200**A to **200**I may be replaced.
(97) According to the second embodiment, the plurality of resonator array structures **200** are integrally formed. As a result, each resonator array structure **200** may be easily installed on the upper wall **111**.
(98) Further, according to the first, second, and fifth embodiments, in the resonator array structures **200**, **200**G to **200**I, some of the cells **220** and **223** surrounded by the plurality of resonators are

formed to the outside of the microwave transmission window. As a result, it is possible to suppress the plasma from spreading horizontally from the metamaterials **200**, **200**G to **200**I.
(99) According to the third and fourth embodiments, the resonator array structures **200**D and **200**F are formed such that the cells **250**, **253**, and **254** surrounded by the plurality of resonators are fitted inside the microwave transmission window. As a result, the plasma density between the cells **250** and **253** can be made uniform.
(100) According to each embodiment, the cells **220** to **223** and **250** to **254** are formed in a grid shape. As a result, high-density plasma can be achieved over a wide range.
(101) Further, according to modifications 3 and 6, in the resonator array structures **200**C and **200**E, the resonant frequency of the outermost resonator of the plurality of resonators (meta-atoms **201**C and **201**E) is a frequency that does not resonate with the microwave. For example, when the resonant frequency fr of other resonators that resonate with the microwave is about 2.35 GHz, the resonant frequency of the outermost resonator of the plurality of resonators is a resonant frequency fr (=about 2.15 GHz or about 2.55 GHz) separated by ±0.2 GHz, for example. As a result, it is possible to suppress interference between the plurality of microwave radiators.
(102) According to each embodiment, the resonator has a size less than 1/10 of the microwave wavelength. As a result, high-density plasma can be achieved over a wide range.
(103) According to the fourth embodiment, the plasma processing apparatus further includes on the upper wall **111**A a gas nozzle (gas introduction nozzle **123**A) that introduces gas for generating the plasma into the processing chamber. In the resonator array structure **200**F, the resonant frequency of the resonator (meta-atom **201**G) on the gas nozzle side is a frequency that does not resonate with the microwave. For example, when the resonant frequency fr of other resonators that resonate with the microwave is approximately 2.35 GHz, the resonant frequency of the resonator on the gas nozzle side is, for example, the resonant frequency fr (=approximately 2.15 GHz or approximately 2.55 GHz) separated by ±0.2 GHz. As a result, it is possible to suppress damage to the gas introduction nozzle **123**A due to the plasma.
(104) According to the fifth embodiment, the resonator array structures **200**G to **200**I have the through hole **214** in the resonator (meta-atom **201**H) adjacent to another cell **223** in the cell **223** surrounded by the plurality of resonators. As a result, the in-plane uniformity of plasma can be improved.
(105) It should be noted that each embodiment disclosed herein is illustrative in all respects but is not restrictive. Each embodiment may be omitted, substituted, or modified in various forms without departing from the scope and spirit of the appended claims.
(106) Although the plasma processing apparatus **100** having the plurality of microwave radiation mechanisms **143** and the metamaterials **200**, **200**A to **200**E, and **200**G to **200**I has been described in the first to third and fifth embodiments, the present disclosure is not limited thereto. For instance, as in the fourth embodiment, the first to third and fifth embodiments may be combined into the plasma processing apparatus having one microwave radiation mechanism **143**.
(107) It should be noted that the present disclosure can also have the following configuration.
(108) (1)
(109) A plasma processing apparatus, comprising: a processing chamber accommodating a substrate, and defining a processing space by an upper wall, a side wall, and a lower wall; a microwave generator configured to generate a microwave for generating plasma; a plurality of microwave radiators provided above the upper wall, and configured to radiate the microwave toward the processing chamber; a plurality of microwave transmission windows provided at positions corresponding to the plurality of microwave radiators in the upper wall, and formed of a dielectric; and a plurality of resonator array structures disposed on lower surfaces of the plurality of microwave transmission windows, respectively, and formed by arranging a plurality of resonators that are capable of resonance with a magnetic field component of the microwave and are smaller in size than a wavelength of the microwave.

(2)
(110) The plasma processing apparatus of (1), wherein each of the plurality of resonator array structures is individually formed.
(111) (3)
(112) The plasma processing apparatus of (1), wherein the plurality of resonator array structures are integrally formed.
(113) (4)
(114) The plasma processing apparatus of any one of (1) to (3), wherein, in the resonator array structure, a portion of cells surrounded by the plurality of resonators extends to an outside of the microwave transmission window.
(115) (5)
(116) The plasma processing apparatus of any one of (1) to (3), wherein the resonator array structure is formed such that cells surrounded by the plurality of resonators are formed to be inside the microwave transmission window.
(117) (6)
(118) The plasma processing apparatus of (4) or (5), wherein the cells are formed in a grid shape.
(119) (7)
(120) The plasma processing apparatus of any one of (1) to (6), wherein, in the resonator array structure, a resonant frequency of an outermost resonator of the plurality of resonators is a frequency that does not resonate with the microwave.
(121) (8)
(122) The plasma processing apparatus of any one of (1) to (7), wherein the resonator has a size less than 1/10 of the wavelength of the microwave.
(123) (9)
(124) The plasma processing apparatus of any one of (1) to (8), further comprising: a gas nozzle formed on the upper wall to introduce a gas for generating the plasma into the processing chamber, wherein, in the resonator array structure, a resonant frequency of a resonator on a gas nozzle side is a frequency that does not resonate with the microwave.
(10)
(125) The plasma processing apparatus of any one of (1) to (9), wherein the resonator array structure comprises a through hole in a resonator adjacent to another cell in a cell surrounded by the plurality of resonators.

Claims

1. A plasma processing apparatus, comprising: a processing chamber accommodating a substrate, and defining a processing space by an upper wall, a side wall, and a lower wall; a microwave generator configured to generate a microwave for generating plasma; a plurality of microwave radiators provided above the upper wall, and configured to radiate the microwave toward the processing chamber; a plurality of microwave transmission windows provided at positions corresponding to the plurality of microwave radiators in the upper wall, and formed of a dielectric; and a plurality of resonator array structures disposed on lower surfaces of the plurality of microwave transmission windows, respectively, and formed by arranging a plurality of resonators that are capable of resonance with a magnetic field component of the microwave and are smaller in size than a wavelength of the microwave.
2. The plasma processing apparatus of claim 1, wherein each of the plurality of resonator array structures is individually formed.
3. The plasma processing apparatus of claim 1, wherein the plurality of resonator array structures are integrally formed.
4. The plasma processing apparatus of claim 1, wherein, in the resonator array structure, a portion

of cells surrounded by the plurality of resonators extends to an outside of the microwave transmission window.
5. The plasma processing apparatus of claim 4, wherein the cells are formed in a grid shape.
6. The plasma processing apparatus of claim 1, wherein the resonator array structure is formed such that cells surrounded by the plurality of resonators are formed to be inside the microwave transmission window.
7. The plasma processing apparatus of claim 6, wherein the cells are formed in a grid shape.
8. The plasma processing apparatus of claim 1, wherein, in the resonator array structure, a resonant frequency of an outermost resonator of the plurality of resonators is a frequency that does not resonate with the microwave.
9. The plasma processing apparatus of claim 1, wherein the resonator has a size less than 1/10 of the wavelength of the microwave.
10. The plasma processing apparatus of claim 1, further comprising: a gas nozzle formed on the upper wall to introduce a gas for generating the plasma into the processing chamber, wherein, in the resonator array structure, a resonant frequency of a resonator on a gas nozzle side is a frequency that does not resonate with the microwave.
11. The plasma processing apparatus of claim 1, wherein the resonator array structure comprises a through hole in a resonator adjacent to another cell in a cell surrounded by the plurality of resonators.
